Hemang Ranga

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.40
SMT. N D MODERN SR SS	Class XII (Rajasthan Board of Secondary Education)	2019	82.40%
SMT. N D MODERN SR SS	Class X (Rajasthan Board of Secondary Education)	2017	79.00%

THESIS

Transaction Foundation Model for Financial Representation Learning

- Reproduced NVIDIA's Transaction Foundation Model (TFM) by training a 29M-parameter **decoder-only Transformer** with self-supervised **causal language modeling** for reusable transaction representations.
- Built an end-to-end **transaction representation learning** pipeline covering financial tokenization, pretraining, embedding extraction, PCA compression, and downstream **XGBoost** evaluation using NVIDIA NeMo.
- Adapted TFM to **IEEE-CIS** through identifier engineering, temporal reconstruction, feature mapping, and chronological splitting; combining TFM embeddings with raw tabular features achieved **14.5% AP improvement** over XGBoost.
- **Ongoing research:** Investigating **schema-robust** transaction foundation models through variable-length field-value serialization and explicit end-of-transaction (**<EOT>**) tokens, with the goal of enabling representation learning and transfer across heterogeneous tabular datasets.

PROJECTS

1) Multilingual Indian Government Schemes RAG Chatbot | *Python, LangChain, ChromaDB, BM25, Gemini API, Ragas, Gradio*

- Architected an end-to-end **hybrid RAG pipeline** combining **BM25, ChromaDB, RRF, and Cross-Encoder reranking**, achieving **86.7% top-5 retrieval hit-rate**.
- Engineered **cross-lingual semantic search** with multilingual embeddings, automated translation, and scheme-aware chunking, achieving **92.4% factual faithfulness** in bilingual responses.
- **Developed a quantitative evaluation suite** utilizing **Ragas** to benchmark Faithfulness, Answer Relevancy, Context Precision, and Context Recall across a 15-question golden dataset, backed by an automated ETL data-indexing pipeline and an interactive **Gradio** web UI.

2) SA-LANDER: Semantic-Adaptive Federated Learning | Python, ML, *DL, Federated learning, Research*

- Extended the **LANDER** (CVPR 2024) framework for Federated Class-Incremental Learning by replacing the fixed bounding radius with an **adaptive per-class radius** via Exponential Moving Average (EMA), enabling the model to dynamically adjust to each class's embedding spread without modifying anchors, architecture, or aggregation.
- Achieved **+7.51% average accuracy** (48.23% to 55.74%) and **-12.82% reduction in catastrophic forgetting** on CIFAR-100 (Non-IID, 5 tasks, 5 clients) vs. paper-reported LANDER; validated **+2.34% accuracy gain** on OrganAMNIST (11-class CT scan) under identical compute budget.
- Implemented adaptive loss B_adaptive = mean(max(0, dist_i r_yi)) with per-class EMA radius clamping; ablation run on Kaggle T4×2 GPU across 80 federated rounds.

3) DNSSEC Tampering Detector | *Python, DNSSEC, BIND9, Docker*

- Built a **DNSSEC validation engine** using Python and dnspython to verify **RRSIG, DNSKEY, DS, KSK, and ZSK** relationships and establish the DNSSEC chain of trust.
- Deployed a containerized **BIND9 DNSSEC environment using Docker** and simulated DNS record tampering by modifying signed records without regenerating cryptographic signatures.
- Implemented automated **DNSSEC integrity verification** to detect **RRSIG signature mismatches and loss of authenticated DNS responses (AD flag)**, followed by restoration of the original DNS configuration.

RELEVANT COURSES

Computer Science: C programming, Data Structures, Algorithms, Discrete Mathematics, Theory of computation, Digital logic design, Computer organization & Architecture, Operating Systems, Computer Networks, Network Security.

Machine Learning: Artificial Intelligence, Foundation of Machine Learning, Deep Learning, Natural Language Processing, Fraud Analysis using machine learning.

Mathematics: Probability & random variables, Statistics, Linear Algebra, Calculus, Linear Optimization.

SKILLS

Programming Languages: C/C++, Python, Java, JavaScript, SQL, HTML, CSS

Machine Learning: PyTorch, NumPy, TensorFlow, scikit-learn, LangChain, HuggingFace Transformers, ChromaDB

Development/Dev Tools: Django, Nodejs, AWS (EC2, RDS, ECS), MySQL, MongoDB, Git, Docker, Linux

POSITIONS OF RESPONSIBILITY

Placement Coordinator - IIT Hyderabad: Selected as a Placement Coordinator responsible for campus placement and internship and handling tasks for Office of Career Services.

Core Member - Prakshepan 4: Managed and lead the team responsible for conducting Prakshepan 4.0, an event for MTech freshers.

President, Inquizitive Club - IIIT Dharwad: Led a 50+ member student club, organising inter-college cultural events.

EXTRACURRICULAR

- Won Bronze medal in hockey for my hostel in inter hostel competition for IITH Milan
- Member of Vibes, Music club of IITH.